3D INDOOR NAVIGATION

USING DEPTH ORDERING

Painter's Algorithm: paint the farthest first, the nearest last.

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Topic 28 | Painter's Algorithm

OBJECTIVE

To implement depth ordering for objects in an indoor navigation environment.

OUTCOME

A working indoor environment with correct rendering of overlapping surfaces.

HOW IT WORKS: FOUR STEPS EVERY FRAME

1 Transform

Convert every polygon (wall, floor, furniture face) to camera space.

2 Measure depth

Distance from the camera to the polygon centre, computed by face_depth().

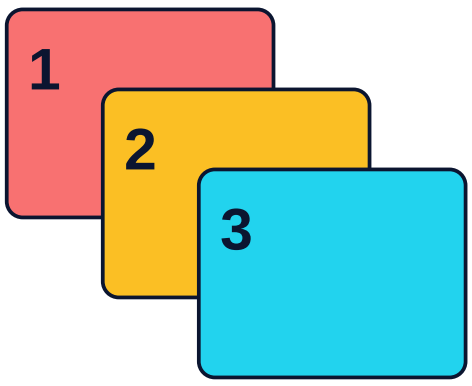
3 Sort

painter_sort() orders all polygons from farthest to nearest.

4 Draw

Paint in that order. Nearer polygons cover farther ones.

THE IDEA: PAINT BACK TO FRONT



Later paint covers earlier paint, so nearer objects correctly hide farther ones. No extra visibility test is needed.

Draw order: 1, 2, 3

THE CORE CODE (painter.py)

```
def face_depth(face, cam):
    cx, cy, cz = centre of the 4 corners
    return (cam.x - cx, EYE - cy, cam.z - cz)

def painter_sort(items):
    # THE PAINTER'S ALGORITHM: farthest first
    return sorted(items, key=lambda it: it[0],
                  reverse=True)

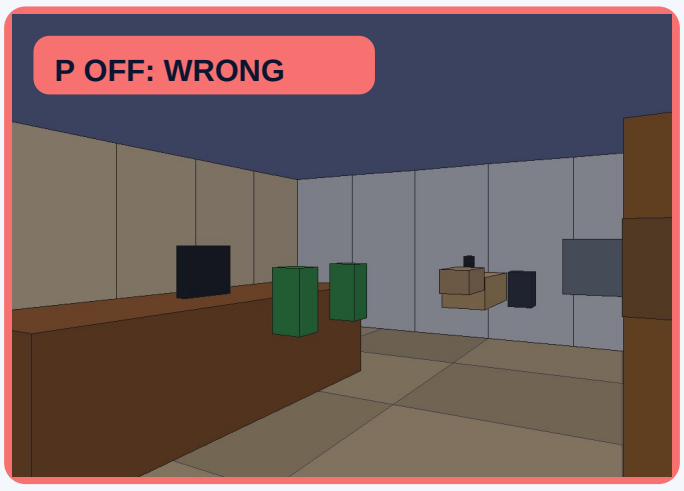
for dist, f, light in items[:count]:  # draw
    draw_face(surface, cam, f, ...)
```

*depth = sqrt(x*x + y*y + z*z) sort key, larger = farther*

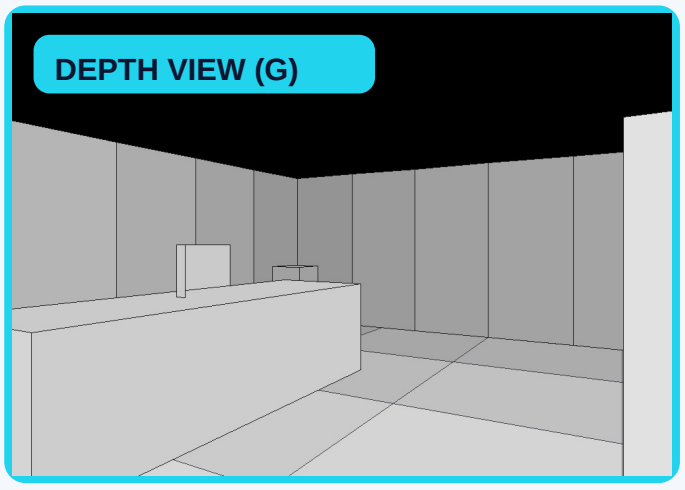
PROOF FROM THE RUNNING PROJECT: SAME VIEW, THREE MODES



Walls and desks hide what is behind them.



Far objects are painted over near ones.



Brighter = nearer, darker = farther.

PROJECT FEATURES

- 4 rooms + corridor with hinged doors that open near you
- Walls split into 1.5 m tiles for accurate sorting
- Back-face culling and near-plane clipping
- Distance fog, collision, live depth labels, minimap

LIMITATIONS AND Z-BUFFER

- Cyclic or intersecting polygons can break the order
- Sorting costs O(n log n) every frame
- Overdraw: hidden polygons are still painted
- A Z-buffer stores depth per pixel, needs no sorting

CONTROLS

W A S D / arrows	move
Q E / mouse drag	turn
P	Painter's on / off
G	depth shading
L	depth labels
- / =	replay draw order